

# Calibrating against Event Mean Concentrations

The EMC comparison method for Observations in OpenHydroQual

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## 1 Overview

Stormwater and BMP monitoring programs often report water quality as *event mean concentrations* (EMCs): one flow-weighted composite concentration per storm, rather than a concentration time series. An `Observation` with `comparison_method = EMC` computes the modeled EMC of every monitored event and compares it with the measured one. The likelihood it returns is used by **Optimize** (genetic algorithm) and **Inverse Run** (MCMC), and by the headless runners OHQ-GA and OHQ-MCMC, the same way as any other observation. Nothing else in the model or in the estimation settings has to change.

## 2 What is computed

For every event  $k$  with window  $[t_{s,k}, t_{e,k}]$ , the modeled EMC is the flow-weighted mean of the observation's expression:

$$\text{EMC}_k = \frac{\int_{t_{s,k}}^{t_{e,k}} C(t) Q(t) dt}{\int_{t_{s,k}}^{t_{e,k}} Q(t) dt}. \quad (1)$$

- $C(t)$  is the observation's `expression` evaluated on its `object`, e.g. `Cu:concentration` on the outlet block.
- $Q(t)$  is the weighting: `emc_weighting_expression` (e.g. `flow`) evaluated on `emc_weighting_object` (e.g. the underdrain link).
- Both are recorded at every accepted solver step. The integrals are trapezoidal over those steps and clipped exactly at the window edges. With  $Q = \text{flow}$ , the numerator is the event's mass load, the denominator its runoff volume, and the ratio is a true flow-weighted EMC.

The **observed EMC** of event  $k$  is the value in `observed_data` whose time stamp lies inside the event window. If several values fall inside, their mean is used.

The **likelihood** is the Gaussian form used by `Least Squared`, applied to the  $N$  events that have both a modeled and an observed EMC (constant terms dropped):

$$-\log L = N \left[ \frac{\text{MSE}}{2\sigma^2} + \log \sigma \right]. \quad (2)$$

- `error_structure = normal`: the residuals are  $\text{EMC}_k^{\text{obs}} - \text{EMC}_k^{\text{mod}}$ , and  $\sigma$  (`error_standard_deviation`) is in concentration units.
- `error_structure = log-normal`: the residuals are  $\log \text{EMC}_k^{\text{obs}} - \log \text{EMC}_k^{\text{mod}}$ , and  $\sigma$  is the standard deviation of the log ratio ( $\sigma = 0.3$  is roughly a  $\pm 35\%$  scatter). Recommended for concentrations.

The fit measures reported per observation (MSE,  $R^2$ , NSE) are computed on the EMCs, in log space when the error structure is log-normal.

## Rules the calculation follows

| Situation                                                    | What happens                                                                                                                 |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| An event has no observed value inside its window             | The event is ignored. It adds nothing to $N$ or to the likelihood.                                                           |
| An event window lies completely outside the simulated period | The event is ignored.                                                                                                        |
| An event window partly overlaps the simulation               | The integrals cover only the overlapping part.                                                                               |
| The modeled weighting integrates to zero over an event       | The EMC is undefined. The time-averaged value of the expression is used instead, and a warning is set on the observation.    |
| No events file, or no weighting values recorded              | The observation contributes zero to the likelihood and sets an error message. Check for this with a forward run (Section 6). |

## 3 Input files

### 3.1 Events file (`emc_events`)

Plain text with two columns per line, `t_start`, `t_end`, in *model time*: the same time base as `simulation_start_time`, `simulation_end_time` and the observed data. Commas, tabs or spaces all work as separators. Lines that do not start with a number (comments, headers) are skipped.

```
// t_start, t_end    (days)
45728.10, 45728.72
45741.30, 45742.05
45760.00, 45760.40
```

- Rows must be sorted by `t_start`; the file is rejected otherwise. `t_end` must be greater than `t_start`.
- Use the **sampling window** of the composite (first to last aliquot) when it is known: that is the period the measured EMC actually represents. Otherwise use the start and end of runoff at the monitoring point.
- Windows should not overlap. If they do, an observed value inside both windows is used for both events.

### 3.2 Observed data (`observed_data`)

The usual two-column time series. Each row is one measured EMC, time-stamped at *any time inside its event window* (e.g. the event midpoint or the start of sampling):

```
45728.40, 18.5
45741.60, 11.2
```

The third event in the events file above has no observed value and is simply skipped. That lets you keep one events file for several analytes that were not all measured in every storm.

## 4 Observation properties

| Property                              | Used by | Meaning                                                                                                                                                |
|---------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>                   | all     | Block or link on which <code>expression</code> is evaluated. For an effluent EMC, the block drained by the outlet, or the outlet link itself.          |
| <code>expression</code>               | all     | The concentration being averaged, e.g. <code>Cu:concentration</code> .                                                                                 |
| <code>observed_data</code>            | all     | Measured EMCs, one per event, time-stamped inside the event window.                                                                                    |
| <code>comparison_method</code>        | all     | Set to <b>EMC</b> .                                                                                                                                    |
| <code>emc_weighting_object</code>     | EMC     | Block or link on which the weighting expression is evaluated, typically the outlet or underdrain link. Leave empty to use <code>object</code> .        |
| <code>emc_weighting_expression</code> | EMC     | The weight; default <code>flow</code> . Use <code>_pos(flow)</code> if the link can reverse, so that backflow does not subtract from the event volume. |
| <code>emc_events</code>               | EMC     | The events file.                                                                                                                                       |
| <code>error_structure</code>          | all     | <b>normal</b> or <b>log-normal</b> .                                                                                                                   |
| <code>error_standard_deviation</code> | all     | $\sigma$ of the residual. It can be assigned to a parameter and estimated.                                                                             |

## 5 Using EMC in the GUI

1. **Build the model** so that the monitored outlet exists as a block and/or link, and the forcing (inflow hydrograph, influent concentration) covers every monitored event. Continuous simulation across the monitoring period is best, because antecedent moisture and accumulated sorbed mass carry over between storms.
2. **Add an Observation** and set, in the property panel:
  - *Object*: the effluent block or outlet link;
  - *Expression*: the concentration, e.g. `Cu:concentration`;
  - *Observed data time series*: the observed-EMC file;
  - *Comparison Method*: `EMC`;
  - *EMC weighting object*: the outlet or underdrain link (or leave empty);
  - *EMC weighting expression*: `flow`;
  - *EMC events file*: the events file;
  - *Error probability distribution*: `log-normal` for concentrations;
  - *Error standard deviation*: an initial guess (e.g. 0.3 for log-normal), or assign a parameter to estimate it.
3. **Run the model** once. The observation's plot shows the *continuous* modeled concentration, not the EMCs; the EMCs are scored after the run. Check that the modeled flow on the weighting object is non-zero during every event.
4. **Calibrate** with *Optimize* (GA) or *Inverse Run* (MCMC) exactly as for any other observation. When they finish, the working folder contains:
  - `fit_measures.txt`: MSE,  $R^2$  and NSE on the EMCs;
  - `mapped_modeled_results.txt`: for EMC observations, the **modeled EMC of each event**, time-stamped like the observed data;
  - the usual GA and MCMC outputs.

Several EMC observations (e.g. dissolved Cu, total Zn and TSS at the same outlet) can share one events file; their negative log-likelihoods add up.

## 6 Using EMC from the console

### 6.1 Script syntax

```
create observation;type=Observation,name=Eff_Cu EMC,object=Outlet_cell,expression=Cu:
concentration,observed_data=eff_Cu_emc.txt,comparison_method=EMC,emc_weighting_object=
Underdrain,emc_weighting_expression=flow,emc_events=events.txt,error_structure=log-normal,
error_standard_deviation=0.3
```

Relative file names are resolved against the model's folder.

### 6.2 Forward run: TOpenHydroQual

```
OpenHydroQual-Console model.ohq -q -r /path/to/OpenHydroQual/resources
```

After the solve, the console scores every observation that has observed data and prints one line per observation:

```
Observation 'Eff_Cu EMC' (EMC): MSE=0.2192 R2=1 NSE=-12.88 -logL=17.31
2 event(s) written to '<workdir>/Eff_Cu EMC EMC.txt'
```

For each EMC observation it also writes <name>\_EMC.txt, with the observed and modeled EMC of every scored event:

```
t, observed EMC, modeled EMC
1.5, 9, 6.22253
5, 7, 4.03986
```

**Always do this forward run before calibrating.** It confirms that the number of scored events is what you expect (too few usually means time stamps outside their windows, or windows outside the simulation), that no **warning:** line appears (zero flow during an event, missing events file, unknown weighting object), and that the modeled EMCs are in a plausible range.

### 6.3 Calibration: OHQ-GA and OHQ-MCMC

```
OHQ-GA    model.ohq [working_folder]
OHQ-MCMC  model.ohq [working_folder]
```

The model's `parameter` entries define what is estimated, and its observations, including EMC ones, define the likelihood. Both runners write `fit_measures.txt` and `mapped_modeled_results.txt` (modeled EMCs per event) in addition to their usual outputs. See `terminal/OHQ-Common/README.md` for all options.

**--kernel (codegen).** A generated kernel records only each observation's expression, not the weighting series. If any observation uses EMC, the runners fall back to the interpreter for the forward solve automatically, so results stay correct but you do not get the kernel speed-up.

## 7 Worked example

A tank receives two storms whose inflow carries a tracer with a first-flush pollutograph. A time-varying outlet link supplies the weighting flow. One parameter, a constant background inflow concentration `p_Cbase`, is calibrated against two observed EMCs. A third event in the events file has no observation and is correctly ignored.

```
create block;type=Reactor,Storage=10,...,name=Tank,time_variable_inflow=Qin.txt,Tracer:
  time_variable_inflow_concentration=Cin.txt,...
create block;type=Reactor,Storage=1000000,...,name=Sink
create link;from=Tank,to=Sink,type=User_flow,flow=Qin.txt,name=Outlet
create parameter;type=Parameter,high=20,low=0.1,name=p_Cbase,prior_distribution=log-normal,
  value=1
setasparameter; object= Tank, parametername= p_Cbase, quantity= Tracer:
  constant_inflow_concentration
create observation;type=Observation,comparison_method=EMC,error_standard_deviation=0.1,
  error_structure=log-normal,expression=Tracer:concentration,name=Effluent EMC,object=Tank,
  observed_data=emc_obs.txt,emc_weighting_object=Outlet,emc_weighting_expression=flow,
  emc_events=events.txt
```

events.txt: 1.0, 2.5 / 4.0, 6.0 / 6.5, 6.8; emc\_obs.txt: 1.5, 9.0 / 5.0, 7.0.

Results:

- The modeled EMCs agree to within about 1% with an independent calculation of  $\int CQ dt / \int Q dt$  from the model's output file.
- **GA** converged to  $p_{Cbase} = 3.97$  (NSE on log-EMCs 0.97).
- **MCMC** gave a posterior median of 3.72, with a 95% interval of [3.14, 5.40].

## 8 Practical notes

- **Which object is the effluent?** For a BMP whose only outlet is an underdrain, weight by the underdrain link's flow and take the concentration in the block that link drains. If an overflow or bypass is *included* in the monitored effluent, the EMC must be computed on the combined outflow: route both paths into one outlet link or node and observe that.
- **Influent EMCs** can be used the same way, e.g. to check a load model: weight by the inflow and observe the inflow concentration.
- **Event definition matters.** A window that is too long dilutes the modeled EMC with baseflow; one that is too short drops the tail of the hydrograph. Match the window to how the composite was actually sampled.
- **Few events, many parameters.** Each event gives one data point. Do not calibrate more parameters than the EMCs can constrain: hold hydraulic parameters from a flow calibration, or share reaction parameters across sites.
- **$\sigma$  and  $N$ .** The weight of an EMC observation in a joint likelihood grows with the number of events  $N$ . When it is combined with a densely sampled time-series observation, estimate each observation's  $\sigma$  rather than fixing them, so that neither data set dominates by sample count alone.